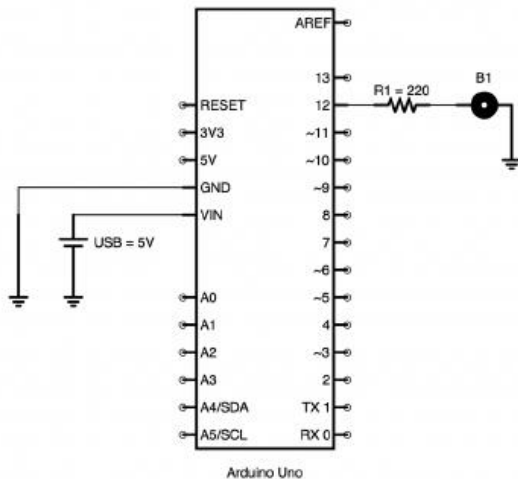


Sub-circuit 1 – Connecting the buzzer

For the second lab we started with adding a buzzer to our circuit , the components we needed were :

- Arduino Uno
- Buzzer
- Resistor
- Breadboard
- Set of male-male jumper wires



We also need to run the test code file in Arduino with the right pin input and the pin in code have to match the pin in board , with the resistor our circuit didn't work and we removed it after that we observe that when we hit the button buzzer makes a sound and if we change delays and HIGH/LOW signals the longevity and the pattern of the sound change .

As we tried to experiment we realized that (buzzerPin, High) and delay(x) are for how long on ring lasts .

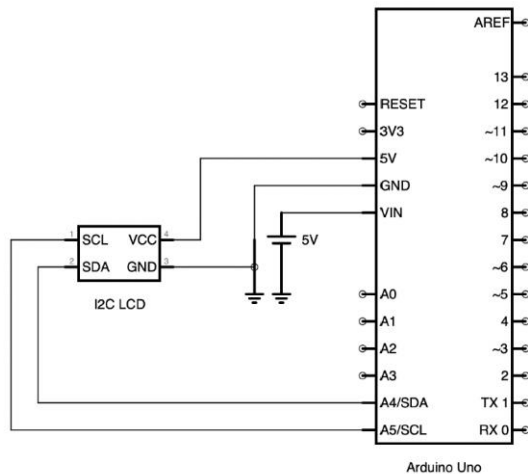
For instance if the delay is 5000 , it means that one ring will last for 5 second .

As for the (buzzerPin , LOW) , this defines that how long a pause between two rings last .

***Image**

***video**

Sub-circuit 2 – Connecting the LCD screen



For task two , we added a LCD screen , and as it was described in the task description we for Arduino we needed to extract I2C_scanner.ino file from the given zip file so that we can find the address of LCD .

As for the board we connected SCL and SDA wires .

```
LiquidCrystal_I2C lcd(0x27, 4, 2); //declaration of the Display - address, signs_per_row, number_of_rows

void setup() {
  // put your setup code here, to run once:
  Serial.begin(9600); //initialisation of serial monitor

  //LCD display initialisation
  lcd.init(); // this commad initiates the LCD
  lcd.backlight(); // if you want the backlight of the LCD, it needs to be initialised
}

void loop() {
  // put your main code here, to run repeatedly:

  lcd.setCursor(0,0); //define the cursor position (within_row, row)
  lcd.print("High"); // print out the desired output, mind the sign limit
  lcd.setCursor(0,1);
  lcd.print("five");
```

```
11:53:26.656 ->
11:53:26.656 -> I2C Scanner
11:53:26.656 -> Scanning...
11:55:39.724 -> I2C device found at address 0x27  !
11:55:39.724 -> done
11:55:39.771 ->
11:55:44.727 -> Scanning...
```

After the address of LCD was scanned the screen turned on and the showing text was “whatever makes sense to show” .

*image of the circuit

After some experiment with screen we realized that the screen only shows 16 characters per row and only two rows we made some changes to it to change the text to “High five” and for that we needed to make the change in (LiquidCrystal_I2C lcd(0x27, 16, 2)) .

#####

```
LiquidCrystal_I2C lcd(0x27, 4, 2); //declaration of the Display - address, signs_per_row, number_of_rows

void setup() {
    // put your setup code here, to run once:
    Serial.begin(9600); //initialisation of serial monitor

    //LCD display initialisation
    lcd.init(); // this command initiates the LCD
    lcd.backlight(); // if you want the backlight of the LCD, it needs to be initialised
}

void loop() {
    // put your main code here, to run repeatedly:

    lcd.setCursor(0,0); //define the cursor position (within_row, row)
    lcd.print("High"); // print out the desired output, mind the sign limit
    lcd.setCursor(0,1);
    lcd.print("five");
```

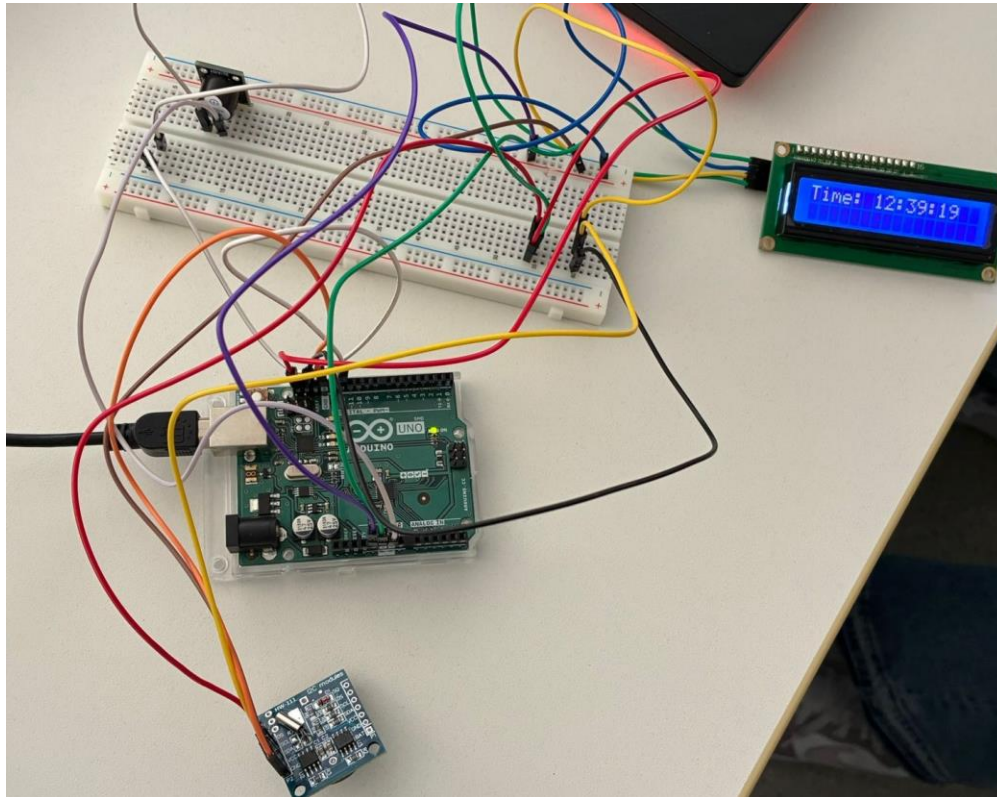
Sub-circuit 3 – Expanding the setup with a Real Time

Clock

The problem that we had at first was that we didn't connect the SDA and SCL wires in series, and we realized that when no address was shown after scanning, after switching to series three addresses were shown.

```
12:21:23.474 -> Scanning...  
12:21:23.474 -> I2C device found at address 0x27 !  
12:21:23.521 -> I2C device found at address 0x50 !  
12:21:23.567 -> I2C device found at address 0x68 !  
12:21:23.614 -> done  
12:21:23.614 ~
```

We set the time with `rtc.adjust()` and Data Time-declaration



Sub-circuit 4 – Using the Push Button

The final task was to add four buttons so for that we went needed to use the test code .

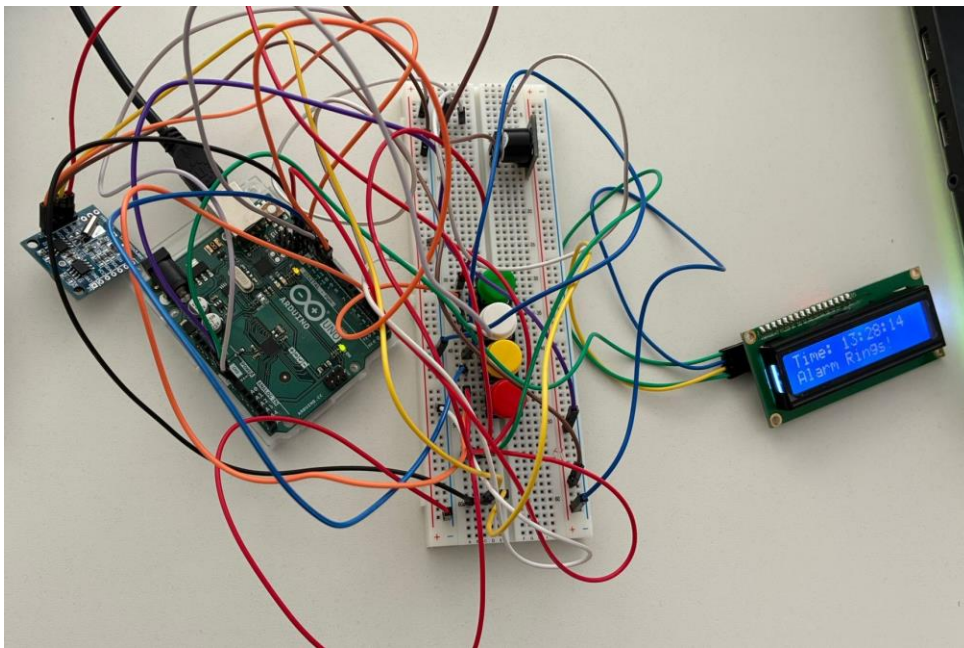
Below i will explain the functionality of each buttons :

Yellow : increase hour in set alarm mode

Green : increase minutes in set alarm mode

White : activation/deactivation of the alarm

Red: set alarm mode



Also the other thing that is good to mention is that we need the male wires to add to the specific pins so all the buttons work properly .